

Rajveer Singh Bharadwaj

Open Source . Compilers . Graphics / Games . Performance .
Algorithms . macOS / iOS

Contact

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Experience

LLVM Compilers (and Subprojects) - LLVM Open Source

Member - LLVM Committer and Compiler Infrastructure Contributor

Self-Employed, Independent Software Engineer | Remote, Worldwide | Sept 2023 - Present

I am an active open source committer to LLVM and its subprojects including llvm, clang, clang-ir, libc++, libc, mlir, aarch64 / x86 / risc-v backend, selection dag, debug-info, inst-combine / vector-combine and many more areas where I have landed and reviewed pull requests closing several issues resolving bugs, compiler crashes and feature additions to the compiler improving performance and optimizing intermediate representation and machine code (ex. llvm-ir, mlir, mir / assembly).

Issues, PRs and Reviews:

- <https://github.com/llvm/llvm-project/issues?q=author:Rajveer100>
- <https://github.com/llvm/clangir/issues?q=author:Rajveer100>
- <https://github.com/llvm/llvm-project/pulls?q=reviewed-by:Rajveer100>
- <https://github.com/llvm/clangir/pulls?q=reviewed-by:Rajveer100>

Swift Compiler (and Subprojects) - Apple Open Source

Contributor - Swift Compiler and Apple Developer Tools

Self-Employed, Independent Software Engineer | Remote, Worldwide | Dec 2022 - Present

As an open source contributor to the Swift compiler and its dependencies covering swift, swift-build, sourcekit-lsp, and upcoming new areas of interest I have fixed errors related to diagnostics, syntax improvement and compiler crashes ranging across swift-frontend, language semantics, expressions, silgen, concurrency, ide and more.

Issues, PRs and Reviews:

- <https://github.com/swiflang/swift/issues?q=author:Rajveer100>
- <https://github.com/swiflang/swift-build/issues?q=author:Rajveer100>
- <https://github.com/swiflang/sourcekit-lsp/issues?q=author:Rajveer100>

Rust Compiler (and Subprojects) - Rust Open Source

Contributor - Rust Compiler and Dependencies

Self-Employed, Independent Software Engineer | Remote, Worldwide | Sept 2023 - Present

I have worked on the rust project covering the core compiler, bootstrap, test-suite, diagnostics, suggestions on unstable features, cleanup as well as dependencies like cargo, file path, areas targeting different platforms and more.

Issues, PRs and Reviews:

- <https://github.com/rust-lang/rust/issues?q=author:Rajveer100>

Bevy Game Engine (and Subprojects) - Bevy Open Source

Contributor - Bevy Game Engine and Dependencies

Self-Employed, Independent Software Engineer | Remote, Worldwide | Nov 2024 - Present

Bevy is an open source game engine written in Rust and based on an ECS (Entity Component System) capable of 2D, 3D graph rendering and a capable animation and sound system. My contributions currently span across its core libraries like ECS, system config, game physics and responsible function/method calls for tracking various actions in animation.

For a detailed list of issues, PRs and reviews checkout:

Issues, PRs and Reviews:

- <https://github.com/bevyengine/bevy/issues?q=author:Rajveer100>

Fidelity Investments - Fidelity Center for Applied Technology

Intern - Fidelity Investments (FCAT Subdivision)

Full Time, Software Engineering Intern | Remote, Bangalore | May 2023 - Jul 2023

As an intern on the FCAT team, I integrated a unified observability solution for Green Meadows databases, visualizing Mongo's Full Time Diagnostic Data (FTDC). Primarily worked with Go, MongoDB, Grafana, Snowflake and Python/Bash for scripting.

Documentation of Work:

- <https://gist.github.com/Rajveer100/488df6d46ba0e5d03293d1320059bb4f>

Education

Bachelor of Technology - Computer Science and Engineering

UnderGraduate - Bangalore Institute of Technology

Student | On-Site, Bangalore | Jun 2020 - June 2024

Achievements

Competitive Programming - ACM ICPC, Google CodeJam, Meta HackerCup

Competitive Programmer - Online Coding Platforms

Remote, Worldwide | Dec 2019 - Sept 2024

- ICPC Regionalist, ACM ICPC Amritapuri Region, Team - '@MainActor' ([Link^](#))
- World Rank 1709 (AIR 389) in Google CodeJam Round 1C
- World Rank 1183 (AIR 857) in Google KickStart Round G
- World Rank 2833 in Meta HackerCup Round 2

Certificates: <https://github.com/Rajveer100/MyCertificates>

Skills

General PL - C++, Swift, Rust, Python, Java, C#

OS - macOS, Terminal, Linux VM (via UTM)

Compiler / Build Tools / Version Control - LLVM, MLIR, CMake, Ninja, Git, NeoVim, Xcode

Game Engines - Unreal Engine (Blueprints/C++), Unity (C#), Bevy

Graphics - Blender 3D

Languages - English (C2), Spanish (A2)

Projects

Compiler Development - LLVM, MLIR, General-Purpose PL

MLIR Toy Compiler, Eva Project (PL), Eva Interpreter, RDP Parser, LLVM Kaleidoscope, Opt Passes

iOS/macOS Development - Swift, SwiftUI, UIKit, Apple Ecosystem

Crypto Tracker, Metal Triangle, AR Model Placer, HotDog Or Not, Flower Classification, Weather Tracker, Flash Chat, AR Ruler

Game Development - Unreal Engine, Blender, Unity, Bevy

SpaceShooter2D (Unity), FPS_Unreal (Unreal Engine), Blender Character Mesh, Flappy_Bird (Bevy)

Full List of Repositories - <https://github.com/Rajveer100?tab=repositories>